ENGN 1580 - Communications Systems

Milan Capoor

Spring 2026

Chapter 1

Fourier Transforms

1.1 Jan 22

1.1.1 Linear Systems

What is a linear system?

1. $L[u+v] = L[u] + L[v]$ (additivity)
2. $L[au] = aL[u]$ (scaling)
3. $L[ax+by] = aL[x] + bL[y]$ (superposition)

Do we actually need scaling if we have additivity?

Example 1.

$$L[x+x] = Lx + Lx = 2Lx$$

seems like we don't. What about rationals?

Example 2. if we want $L[0.1x]$ we can see

$$L[x] = L[0.1x + \cdots + 0.1x] = 10L[0.1x] \implies L[0.1x] = \frac{1}{10}L[x]$$

still working...

Example 3. And even for negatives...

$$L[-x] = L[x - x - x] = L[x] + 2L[-x]0 = L[x] + L[-x]L[-x] = -L[x]$$

(**Remark:** this also immediately leads to the identity $L[0] = 0$ for all linear systems)

Example 4 (A counter): What if $L[ax]$ is irrational? We can approximate by rationals and apply the same arguments above but this requires L continuous on $\mathbb{R} \setminus \mathbb{Q}$ – a rather strong condition

Where things really break are $\mathbb{C}$: what could we possibly say about $L[jx]$? (convention: j is imaginary basis in this class)

1.1.2 Periodic Signals

Signal: a function

periodic signals: from fourier series

$$p(t) = \sum p_k exp(2\pi jk/T)$$

for period T where

$$p_k = \frac{1}{T}\int_{-T/2}^{T/2} p(t)e^{-j*2\pi/T*kt}\,dt$$

(as an aside notice that this is a projection)

Fourier transform:

$$\mathfrak{X}f = \int x(t)e^{-j*2\pi ft}\,dt$$

Inverse Fourier:

$$x(t) = \int \mathfrak{X}fe^{j*2\pi ft}\,df$$

where f is the frequency (note we are missing the coefficients because we work in Hz)

Laplace:

$$x(s) = \int x(t)e^{-st}\,dt$$

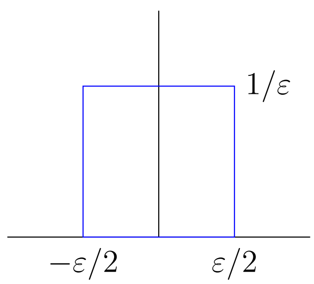
Notice that all of these transforms have integrals of exponentials. Why might that be?

Linear time invariant (LTI) systems: Let L[x(t)] = y(t). If this is an LTI,

$$L[x(t-\tau)] = y(t-\tau)$$

impulse: square function of lengths $1/\varepsilon$ centered at 0 satisfying

1. $\int \delta(t)\,dt = 1$
2. $\int \delta(t-a)s(t)\,dt = s(a)$(sifting property)



This leads to an alternative definition of an LTI $h(t)$ as one for which

1. $h(\delta(t)) = h(t)$
2. $h(a\delta(t-\tau)) = ah(t-\tau)$

Let us consider a general case

$$x(t) = \int x(\tau)\delta(t-\tau)d\tau$$

now what is $h(x(t))$?

If we view the integral as a Riemann sum, we can see easily that

$$\begin{aligned} h(x(t)) &= h(\int x(\tau)\delta(t-\tau)d\tau) \\ &= \int x(\tau)h(t-\tau)d\tau \\ &= y(t) \end{aligned}$$

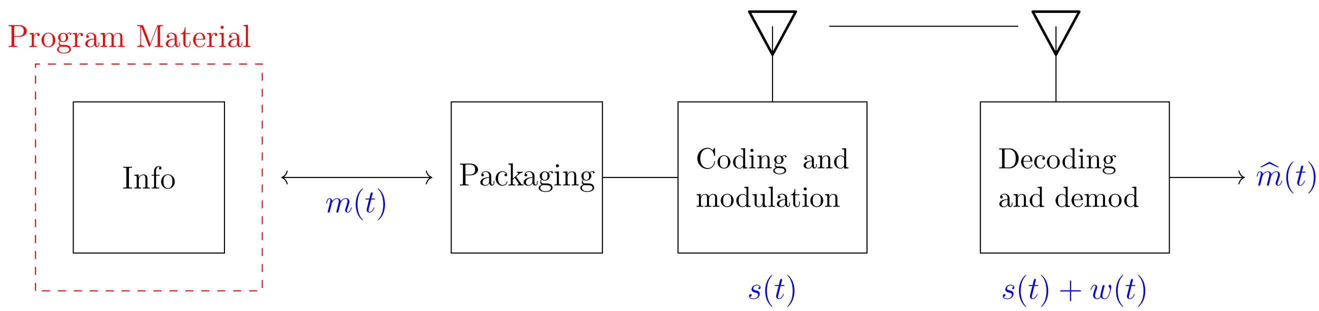
a convolution! (and of course we could equally see by a change of variables $\int x(t-\tau)h(\tau)d\tau$)

What about $h(e^{jst})$?

$$\int e^{js(t-\tau)}h(\tau)d\tau = e^{jst}\int h(t)e^{-jst}d\tau$$

1.2 Jan 27

Outline of a Communications System:



where:

- info is viewed broadly as songs, video, text, etc.
- $m(t)$ is the message signal (assuming no loss between program material and packaging)
- $s(t)$ is the transmitted signal
- $w(t)$ is noise added by the transmission interference
- $\widehat{m}(t)$ is the received (estimated) message signal

1.2.1 Linear system review:

A system L is **linear** if it satisfies *superposition*:

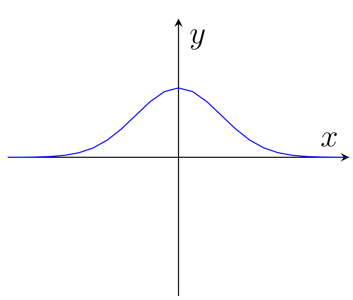
$$Lax + by = aL[x] + bL[y]$$

If we define a delta function $\delta(t)$ satisfying

1. $\int_{-\infty}^{\infty} \delta(t)\, dt = \int_{0^-}^{0^+} \delta(t)\, dt = 1$
2. $\int \delta(t-a)s(t)\, dt = s(a)$ (sifting property)

Last time we saw the square pulse. However, we can also define it as the limit of a Gaussian as $\sigma \to 0$:

$$\delta_\sigma(t) = \sqrt{\frac{1}{2\pi\sigma^2}}e^{-\frac{t^2}{2\sigma^2}}$$



> **Exercise:** Show that $\delta_\sigma(t)$ is a delta function as $\sigma \to 0$.

Impulse response of a linear system:

$$\delta(t - t_0) \longrightarrow \boxed{\quad h(t) \quad} \longrightarrow h(t - t_0)$$

With a little work, we can generalize this property to see that for any input signal $x(t)$, the output of a linear system is given by the convolution of the input with the impulse response:

$$\begin{aligned} y(t) &= h(x(t)) \\ &= \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\, d\tau \\ &= \int_{-\infty}^{\infty} x(t-\tau)h(\tau)\, d\tau \end{aligned}$$

Fourier Transform:

$$h(t) \overset{\mathcal{F}}{\Longrightarrow} H(f) = \int_{-\infty}^{\infty} h(t)e^{-j\,2\pi f t}\, dt$$

Example: Let $x(t) = \cos 2\pi f_0 t$. What is $X(f)$?

$$\begin{aligned} \cos 2\pi f_0 t &= \frac{e^{j\,2\pi f_0 t} + e^{-j\,2\pi f_0 t}}{2} \\ h(\cos 2\pi f_0 t) &= \frac{1}{2}h(e^{j\,2\pi f_0 t}) + \frac{1}{2}h(e^{-j\,2\pi f_0 t}) \\ &= \frac{1}{2}e^{j\,2\pi f_0 t}H(f_0) + \frac{1}{2}e^{-j\,2\pi f_0 t}H(-f_0) \end{aligned}$$

Let's generalize:

$$\begin{aligned} h(x(t)) = y(t) - &\int x(\tau)h(t-\tau)\, d\tau \\ &\overset{\mathcal{F}}{=} \int\int x(\tau)h(t-\tau)e^{-j\,2\pi f t}\, dt\, d\tau \\ &= \int x(\tau)\left(\int h(t-\tau)e^{-j\,2\pi f t}\, dt\right) d\tau \\ &= \int x(t)\left[\int h(u)e^{-j\,2\pi f(u+\tau)}\right] d\tau \qquad (u = t - \tau) \\ &= \int x(t)\underbrace{\left[\int h(u)e^{-j\,2\pi f u}\right]}_{H(f)} e^{-j\,2\pi f \tau}\, d\tau \\ &\overset{\mathcal{F}^{-1}}{=} X(f)H(f) \end{aligned}$$

Which yields the important result: *convolution in the time domain is multiplication in the frequency domain:*

$$x(t) \star h(t) = (x \star h)(t) \overset{\mathcal{F}}{=} X(f)H(f)$$

In fact, the work here leads to a number of useful properties of the Fourier transform:

Time shift property:

$$\boxed{\mathcal{F}[x(t-t_0)] = e^{j\,2\pi f t_0} X(f)}$$

Scaling:

$$\begin{aligned}\mathcal{F}[x(at)] &= \int x(at) e^{-j\,2\pi f t}\,dt \\ &= \int x(u) e^{-j\,2\pi f (u/a)} \frac{du}{a} \qquad (du = a\,dt) \\ &= \boxed{\frac{1}{|a|} X(f/a)}\end{aligned}$$

Differentiation:

$$\begin{aligned}\frac{d}{dt}x(t) &= \frac{d}{dt}\left[\int X(f) e^{j\,2\pi f t}\,df\right] \\ &= \int X(f)(j\,2\pi f) e^{j\,2\pi f t}\,dt\end{aligned}$$

so

$$\boxed{\mathcal{F}\left[\frac{dx}{dt}\right] = X(f)\;j2\pi f}$$

Integration:

$$\boxed{\mathcal{F}\left[\int_{-\infty}^{t} x(u)\,du\right] = \frac{X(f)}{j\,2\pi f} + \frac{X(0)}{2}\delta(f)}$$

Exercise: Prove the integration property

Parseval:

$$\boxed{\int_{-\infty}^{\infty} x^2(t) = \mathcal{E}_x = \int_{-\infty}^{\infty} |X(f)|^2\,df}$$

("the energy in time domain = energy in frequency domain")

We've seen over and over that $\mathcal{F}[x(t)] = X(f)$? What about $\mathcal{F}[X(f)]$?

First, two fact that should be obvious:

$$\begin{aligned}\mathcal{F}[\delta(t)] &= 1 \\ \mathcal{F}[\delta(t-\tau)] &= e^{-j\,2\pi f t}\end{aligned}$$

and since (up to sets of measure zero),

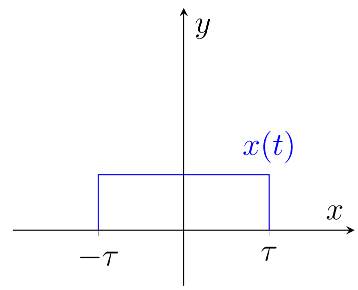
$$\delta(t-\tau) \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\Longleftrightarrow}} e^{-j\,2\pi f t}$$

which gives us

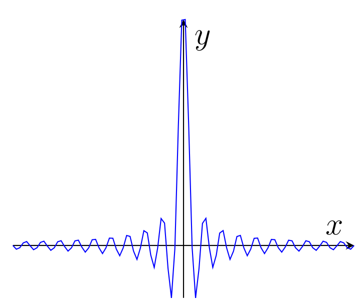
Duality:

$$\begin{aligned}
\mathcal{F}[X(t)] &= \int X(t)e^{-j\,2\pi ft}\,dt \\
&= \int x(u)e^{-j\,2\pi(-tu)}\,du \\
&= \int\int x(u)e^{-j\,2\pi tu}e^{-j\,2\pi t}\,du\,dt \\
&= \int x(u)\left[\int e^{-j\,2\pi t(u+f)}\,dt\right]\,du \\
&= \int x(u)\delta(u+f)\,du \\
&= x(-f)
\end{aligned}$$

Graphically, for a square function,

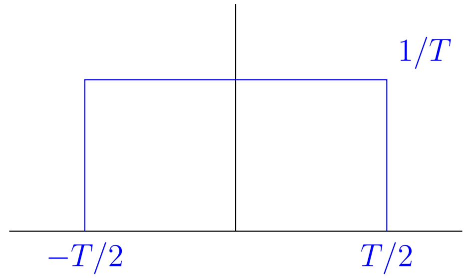


$$\begin{aligned}
\mathcal{F}[x(t)] &= \int_{-T}^{T} e^{-j\,2\pi ft}\,dt \\
&= \left(\frac{1}{-j\,2\pi f}\right)e^{-j\,2\pi ft}\Big|_{-T}^{T} \\
&= -\frac{1}{j\,2\pi f}\left(e^{-j\,2\pi fT}-e^{j\,2\pi fT}\right) \\
&= -\frac{1}{\pi f}\sin(-2\pi fT) \qquad \left(\sin z = \frac{e^{jz}-e^{-jz}}{2j}\right) \\
&= \frac{\sin 2\pi fT}{\pi f}
\end{aligned}$$



1.3 Jan 29

Recall: What is the Fourier transform of this box?



(Note this is an impulse as $T \to 0$)

$$\mathcal{F} = \int_{-T/2}^{T/2} \frac{1}{T} e^{-j\,2\pi f t}\,dt = \frac{1}{T}\left(\frac{1}{-j\,2\pi f}\right) e^{-j\,2\pi f t}\Big|_{-T/2}^{T/2} = \frac{1}{-j\,2\pi f T}\left(e^{-j\,\pi f T} - e^{j\,\pi f T}\right) = \frac{\sin(\pi f T)}{\pi f T}$$

How might we sanity check this?

We already saw that as $T \to 0$, this becomes an impulse in the time domain. What happens in the frequency domain?

$$\lim_{T\to 0} \frac{\sin \pi f T}{\pi f T} = 1$$

and indeed this matches the rule $\mathcal{F}(\delta(t)) = 1$.

Example:

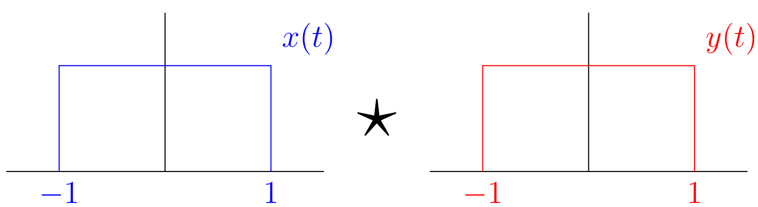
$$\mathcal{F}[e^{-at}u(t)] = \int_0^\infty e^{-at} e^{-j\,2\pi f t}\,dt = \int_0^\infty e^{-(a+j\,2\pi f)t}\,dt = \frac{1}{a + j\,2\pi f}$$

(for $a > 0$ and $u(t)$ the unit step function)

Convolution:

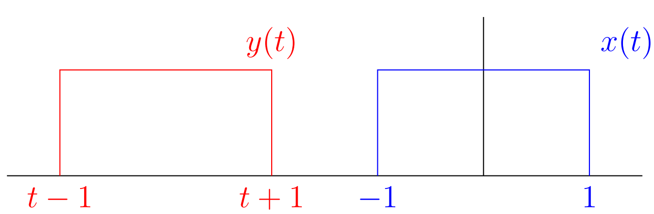
$$\int_{-\infty}^{\infty} x(\tau) y(t-\tau)\,d\tau$$

Example: Convolve two box functions of width 2:

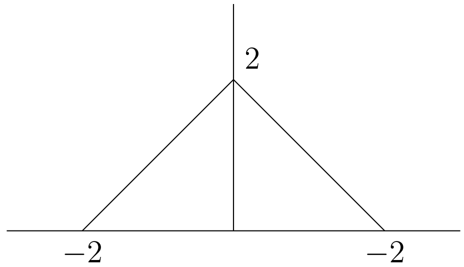


We could do this directly or we can think about it geometrically.

$y(t-\tau)$ corresponds to a reflection and shift by t.

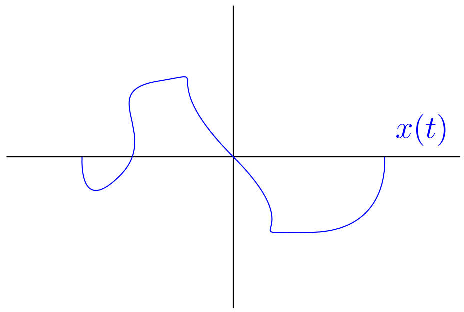


Clearly, multiplying these two functions will give nonzero output only when they overlap. When do they overlap? Precisely at $t \in (-2, 2)$.



Notice that the length of the domain was 2 before convolving and 4 after convolving. This makes sense since convolution is a "spreading and smoothing" operation

Example: $\delta(t) \star x(t)$ where $x(t)$

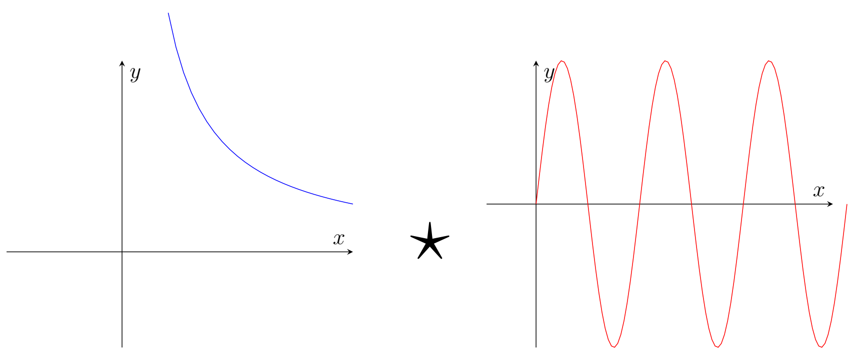


Trick question!

$$\int \delta(x-\tau)x(\tau)\, d\tau = x(t)$$

(sifting!)

For this class, we should think about convolutions critically and graphically:



What is the convolution of these two functions at $t = -2$?

These would be quite annoying to do directly but if we notice that $x(t)$ is undefined for $t \leq 0$ then we can see that there is no overlap at all and the convolution is 0.

Recall:

$$\mathcal{F}(x(t)) = X(f)$$
$$\mathcal{F}(X(t)) = x(-f)$$
$$\mathcal{F}(\delta(t)) = 1$$

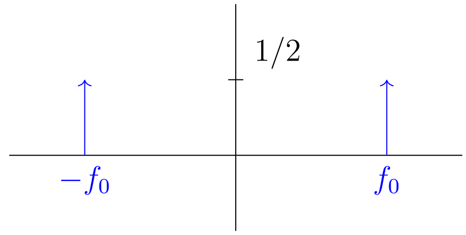
so

$$\mathcal{F}(\delta(t-t_0)) = e^{-j\,2\pi f t_0}$$
$$\mathcal{F}(e^{-j\,2\pi f_0 t}) = \delta(f+f_0)$$

Example: $\mathcal{F}(\cos 2\pi f_0 t)$ does not converge analytically BUT

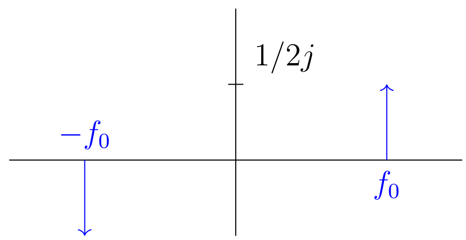
$$\cos 2\pi f_0 t = \frac{e^{j\,2\pi f_0 t} + e^{-j\,2\pi f_0 t}}{2}$$

so from the above,



Similarly,

$$\begin{aligned}\mathcal{F}(\sin 2\pi f_0 t) &= \mathcal{F}\left(\frac{1}{2j} e^{j\,2\pi f_0 t} - e^{-j\,2\pi f_0 t}\right) \\ &= \frac{1}{2j}(\delta(f + f_0) - \delta(f - f_0))\end{aligned}$$



Complex conjugation: $(a + bj)^* = a - bj$

We know

$$\mathcal{F}(x(t)) = \int x(t) e^{-j\,2\pi f t}\,dt = X(f)$$

so what is $\mathcal{F}(x^*(t))$?

$$\begin{aligned}F(x^*(t)) &= \int x^*(t) e^{j\,2\pi f t}\,dt \\ &= \int x^*(t) e^{-j\,2\pi(-f)t}\,dt \\ &= X(-f) \\ &= X^*(f)\end{aligned}$$

leading to the property:

$$\boxed{X^*(f) = X(-f)}$$

Even/Odd Property: Let $x(t) = x(-t)$. Then

$$\int_{-\infty}^{\infty} x(t) e^{-j\,2\pi f t}\,dt = \int_{0}^{\infty} x(t) e^{-j\,2\pi f t} + x(t) e^{j\,2\pi f t}\,dt = 2\int_{0}^{\infty} x(t) \cos 2\pi f t\,dt \in \mathbb{R}$$

Even functions have real fourier transforms. Odd functions have imaginary fourier transforms.

1.4 Linear Systems Assessment

Exercise: The Fourier transform of $x(t)$ is $X(f)$. Prove that the Fourier transform of $x(t - t_0)$ is $e^{-j\,2\pi f t_0} X(f)$.

$$
\begin{aligned}
\mathcal{F}(x(t-t_0)) &= \int x(t-t_0) e^{-j\,2\pi f t}\,dt \\
&= \int x(u) e^{-j\,2\pi f (u+t_0)}\,du \\
&= e^{-j\,2\pi f t_0} \int x(u) e^{-j\,2\pi f u}\,du \\
&= e^{-j\,2\pi f t_0} X(f)
\end{aligned}
$$

Exercise: The energy in a signal $x(t)$ is defined as

$$E = \int_{-\infty}^{\infty} |x(t)|^2\,dt$$

Prove that if $X(f)$ is the Fourier transform of $x(t)$, then we must also have

$$E = \int_{-\infty}^{\infty} |X(f)|^2\,df$$

Hint: You may use the fact that

$$\int_{-\infty}^{\infty} e^{j\,2\pi v t}\,dt = \delta(v)$$

$$
\begin{aligned}
|x(t)|^2 &= x(t)x^*(t) \\
x(t) &= \int X(f) e^{j\,2\pi f t}\,df \\
x^*(t) &= \int X^*(f) e^{-j\,2\pi f t}\,df \\
E &= \int |x(t)|^2\,dt \\
&= \int \left(\int X(f) e^{j\,2\pi f t}\,df \right) \left(\int X^*(f') e^{-j\,2\pi f' t}\,df' \right) dt \\
&= \int \int \int X(f) X^*(f') e^{j\,2\pi t (f-f')}\,df\,df'\,dt \\
&= \int \int X(f) X^*(f') \left(\int e^{j\,2\pi t (f-f')}\,dt \right) df\,df' \\
&= \int \int X(f) X^*(f') \delta(f-f')\,df\,df' \\
&= \int X(f) X^*(f)\,df \\
&= \int |X(f)|^2\,df
\end{aligned}
$$

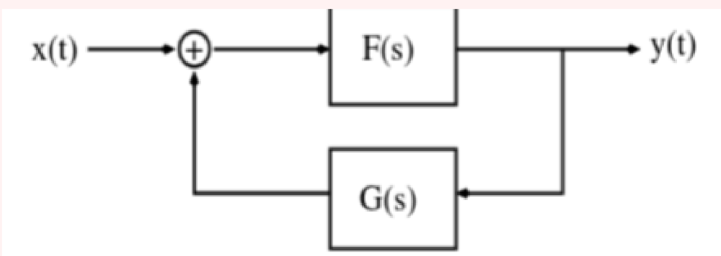
Exercise: A signal $x(t)$ of finite energy is applied to a square-law device whose output is defined by $y(t) = x^2(t)$. The spectrum $X(f)$ of $x(t)$ is limited to the frequency interval $-W \le f \le W$. Show that the spectrum, $Y(f)$ of $y(t)$ is limited to the frequency interval $-2W \le f \le 2W$.

$$
\begin{aligned}
y(t) &= x^2(t) \\
\mathcal{F}(y(t)) &= Y(f) = \mathcal{F}(x(t)x(t)) \\
&= X(f) \star X(f) \\
&= \int_{-\infty}^{\infty} X(\tau)X(f-\tau)\, d\tau \\
&= \int_{-W}^{W} X(\tau)X(f-\tau)\, d\tau \qquad \text{(since } X(f) = 0 \text{ for } |f| > W)
\end{aligned}
$$

At the limit $\tau = -W$, $-W \le f \le W \implies f \in [0, 2W]$. A symmetric argument holds for $\tau = W$ so $Y(f) = 0$ for $|f| > 2W$.

Exercise: Show that the overall transfer function $H(s)$ for the feedback system below is given by

$$H(s) = \frac{F(s)}{1 - F(s)G(s)}$$



Let $W(s)$ be the Laplace transform of the input to $F(s)$.

Then

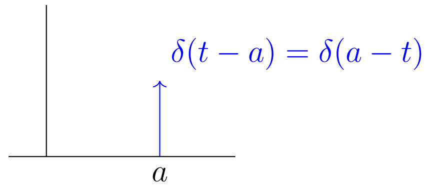
$$
\begin{aligned}
W(s) &= X(S) + G(s)Y(s) \\
Y(s) &= F(s)W(s) \\
&= F(s)\left[X(s) + G(s)Y(s)\right] \\
H(s) &= \frac{Y(s)}{X(s)} \\
&= \frac{F(s)\left[X(s) + G(s)Y(s)\right]}{X(s)} \\
&= F(s)\left[1 + G(s)H(s)\right]
\end{aligned}
$$

so

$$
\begin{aligned}
H(s) - F(s)G(s)H(s) &= F(s) \\
H(s)\left[1 - F(s)G(s)\right] &= F(s) \\
\implies H(s) &= \frac{F(s)}{1 - F(s)G(s)}
\end{aligned}
$$

1.5 Quizlette 1

Exercise: Sketch $s_1(t) = \delta(t-a)$ and $s_2(t) = \delta(a-t)$. What are the respective outputs if $s_1(t)$ and $s_2(t)$ are applied to the input of a linear time invariant system with impulse response $h(t)$?



Recall: the impulse response satisfies

$$\begin{aligned}
\int h(\tau)\delta(t-a-\tau)\, d\tau &= h(t-a) \\
\int h(\tau)\delta(a-t-\tau)\, d\tau &= \int h(\tau)\delta(t-a-\tau)\, d\tau \\
&= h(t-a)
\end{aligned}$$

Exercise: the impulse response of a system is $e^{-t}u(t)$. What is the system output if the input is $e^{-2t}u(t)$? Show your work

The output of a linear system if the convolution of the input with the impulse response.

$$\begin{aligned}
y(t) &= e^{-t}u(t) \star e^{-2t}u(t) \\
&= \int e^{-\tau}u(\tau) \cdot e^{-2(t-\tau)}u(t-\tau)\, d\tau \\
&= \int_0^t e^{-\tau}e^{-2(t-\tau)}\, d\tau \\
&= e^{-2t}\int_0^t e^{\tau}\, d\tau \\
&= e^{-t} - e^{-2t}
\end{aligned}$$

Exercise: Suppose

$$x(t) = \left(\frac{\sin 2\pi t}{\pi t}\right)^2$$

What is $X(f)$?

We know that

$$R = \mathcal{F}\left(\frac{\sin 2\pi t}{\pi t}\right) = \frac{1}{2j\,\pi t}[\delta(f-1) + \delta(f+1)]$$

so

$$\begin{aligned}
\mathcal{F}(R^2) &= R \star R \\
&= \int R(\tau)R(f-\tau)\, d\tau \\
&= \int \frac{1}{2j\,\pi\tau}[\delta(\tau-1)+\delta(\tau+1)] \cdot \frac{1}{2j\,\pi(f-\tau)}[\delta(f-\tau-1)+\delta(f-\tau+1)]\, d\tau \\
&= -\frac{1}{(2\pi)^2}\int \frac{1}{\tau(f-\tau)}[\delta(\tau-1)+\delta(\tau+1)][\delta(f-\tau-1)+\delta(f-\tau+1)]\, d\tau
\end{aligned}$$

Exercise: Suppose $y(t) = x(t)\cos 2\pi f_0 t$. Is the system $y(t) = L[x(t)]$ linear? Is it time-invariant?

$$\begin{aligned}
L[a_1x_1(t) + a_2x_2(t)] &= (a_1x_1(t) + a_2x_2(t))\cos 2\pi f_0 t \\
&= a_1x_1(t)\cos 2\pi f_0 t + a_2x_2(t)\cos 2\pi f_0 t \\
&= a_1L[x_1(t)] + a_2L[x_2(t)]
\end{aligned}$$

So linear.

Now, time-invariance:

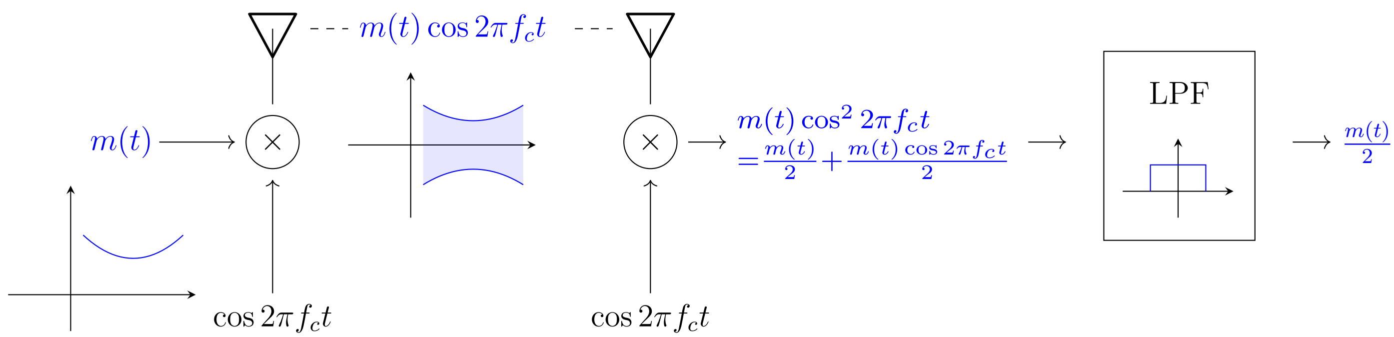
$$\begin{aligned}
L[x(t-t_0)] &= x(t-t_0)\cos 2\pi f_0 t \\
&\neq y(t-t_0) = x(t-t_0)\cos 2\pi f_0(t-t_0)
\end{aligned}$$

so not time-invariant.

Chapter 2

Modulation

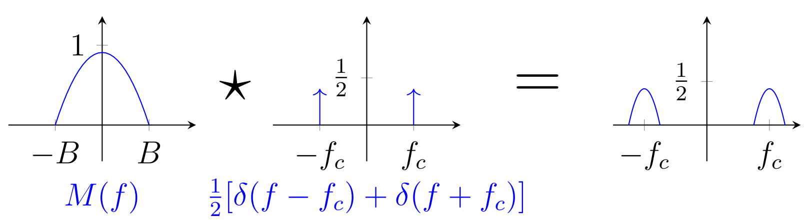
Consider a simple modulation and demodulation scheme:



for $m(t)$ a message and f_c a carrier frequency ($f_c \gg B$, the bandwidth of $m(t)$) which transmits the message better since it is higher frequency.

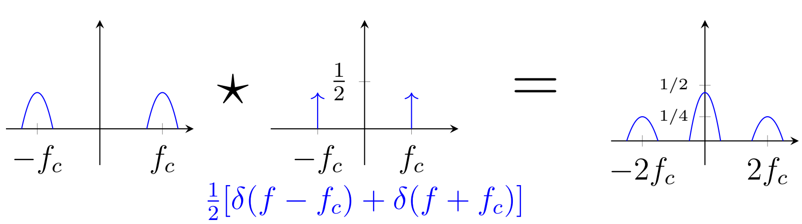
What exactly is happening here? Let's look in the frequency domain:

1. Modulation:



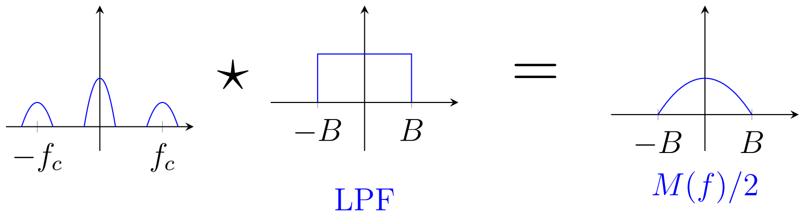
   Convolving with an impulse shifts the signal to be centered at the impulse!

2. Demodulation:



   (spread a copy of the signal around $-f_c$ and f_c so that the interference terms reinforce each other at the baseband)

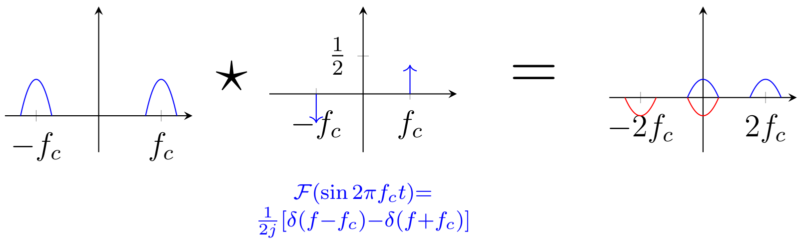
3. Filtering:



2.1 Feb 3

Coherent AM: Synchronous modulation; Modulate with $\cos 2\pi f_c t$ and demodulate with the same signal.

Recall in our demodulation step we multiplied by $\cos 2\pi f_c t$ again. What if we made a mistake and used $\sin(2\pi f_c t)$ instead?

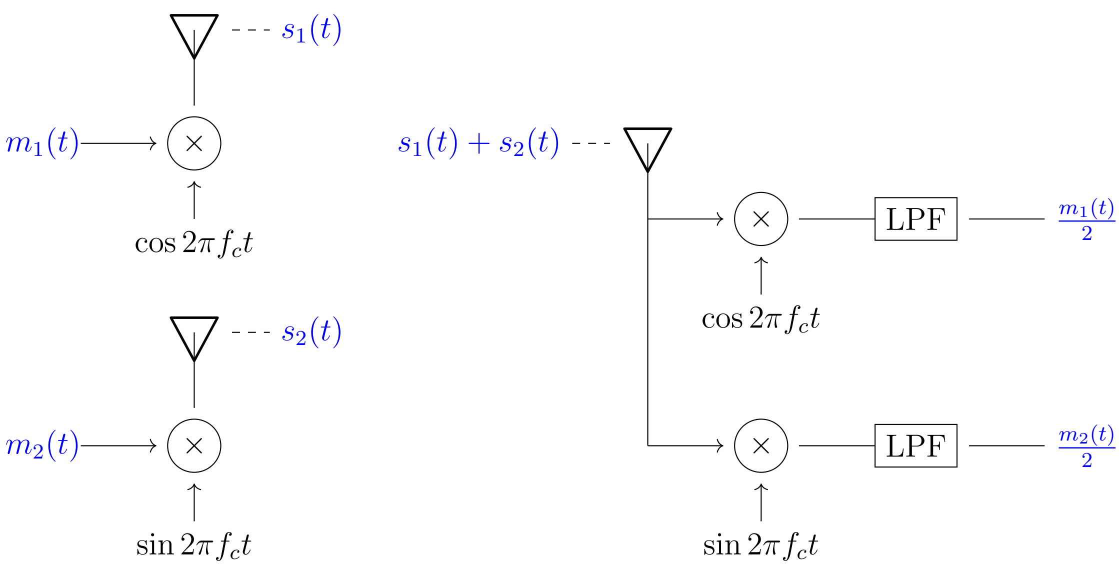


So the signals cancel around the baseband and we get no message after filtering!

How could we recover if we made this mistake?

Let's try broadcasting two messages $m_1(t)\cos 2\pi f_c t$ and $m_2(t)\sin 2\pi f_c t$ simultaneously.

The received signal is $(s_1(t) + s_2(t))$. If we demodulate by $\cos 2\pi f_c t$ on one rail and $\sin 2\pi f_c t$ on the other rail and filter, we get two separate messages back, $m_1(t)/2$ and $m_2(t)/2$ respectively!



Single sideband modulation: Let $m(t)$ be a real signal with Fourier transform $M(f)$. We know

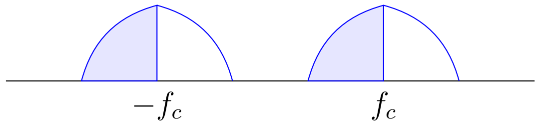
$$M(f) = \int m(t) e^{-j\,2\pi f t}\,dt$$

$$M^*(f) = \int m(t) e^{-j\,2\pi(-f)t}\,dt = M(-f)$$

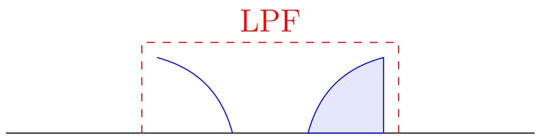
(i.e. the real part is even and the imaginary part is odd)

In particular, this means we can send less information by sending only the positive frequencies or only the negative frequencies and reconstructing the other half using the even/odd properties.

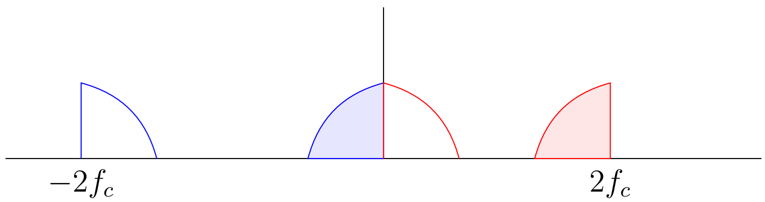
1. Modulate $m(t)$ by $\cos 2\pi f_c t$ to shift the spectrum to be centered at $\pm f_c$.



2. Filter



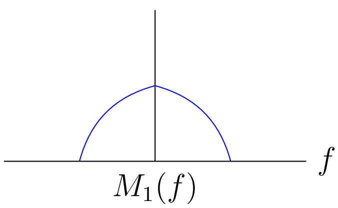
3. Transmit
4. Demodulate by multiplying by $\cos 2\pi f_c t$ again.

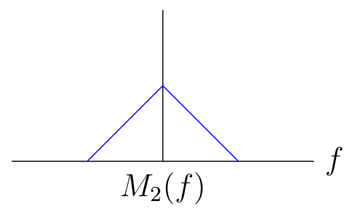


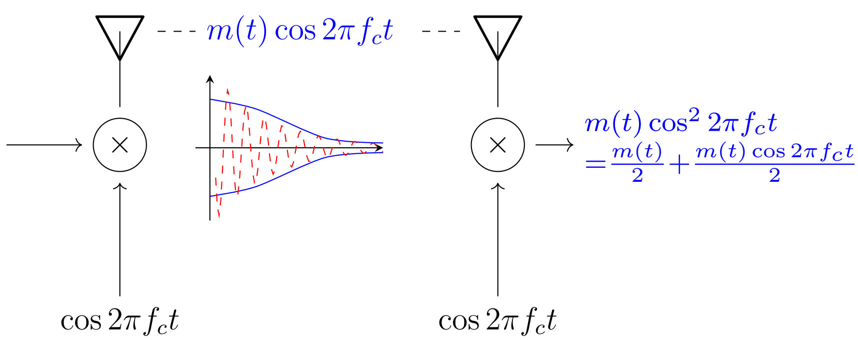
5. LPF again to recover the full signal using half the bandwidth!

Frequency Division Multiplexing:

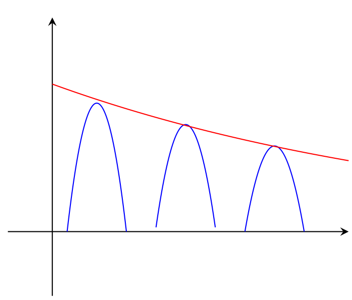
Define two signals $m_1(t)$ and $m_2(t)$.



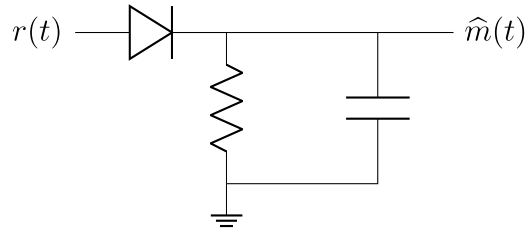




Though we could also blow up in time and rectify the transmitted wave



so we can recover the envelope (all the information is in the amplitude of the envelope), for example using this circuit:



Motivating FM:

$$r(t) = \sin\left(2\pi f_c t + \int_{-\infty}^{t} m(\tau) d\tau\right)$$

Where is the information here? In the integral.

What does it look like? First, we know it has constant amplitude ("modulus 1").

Define the *instantaneous frequency*

$$\omega(t) = \frac{d}{dt}\Theta(t) = 2\pi f_0 = \omega_0$$

for $\Theta(t) = 2\pi f_0 t$ the *instantaneous phase* of the signal.

Now let's try differentiating:

$$\frac{d}{dt} r(t) = (2\pi f_c + m(t)) \cos\left(2\pi f_c t + \int_{-\infty}^{t} m(\tau)\, d\tau\right)$$

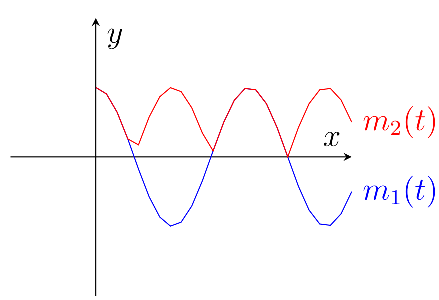
2.2 Feb 5

2.2.1 AM Review

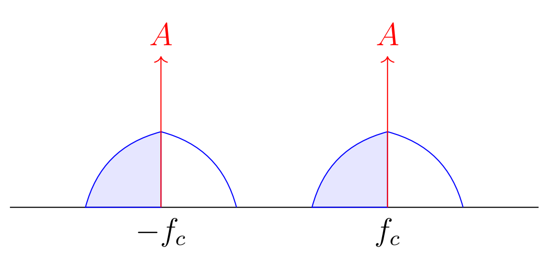
Coherent Modulation: Modulate and demodulate by multiplying with the same carrier signal (traditionally $\cos 2\pi f_c t$). Recover $\alpha m(t)$ after low-pass filtering.

Large Carrier (LC) Modulation: Transmit the message $(m(t) + A)\cos 2\pi f_c t$ with $A \geq \max_t |m(t)|$ (though this is a bit overkill to ensure $m(t) + A \geq 0$). Receive with an envelope detector, LPF, and very low HPF (to remove the DC offset A).

Why do we need to make sure the message is always positive? To an envelope detector, there is no difference between $m(t) + A$ and $|m(t) + A|$. If $m(t) + A$ goes negative, the envelope detector will just flip it positive and distort the message.

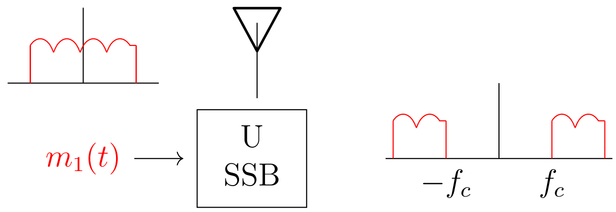


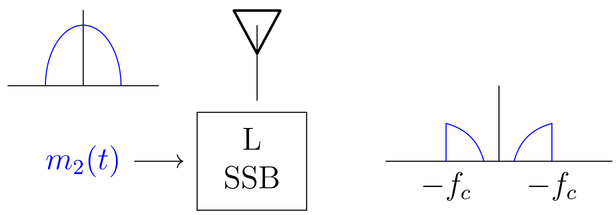
What does the transmitted signal look like in the frequency domain?



Exactly the same as coherent AM but with impulses added at the message frequencies.

Single Sideband (SSB) Modulation: can be coherent or incoherent. Before transmitting apply an LPF so we use half the bandwidth. This takes advantage of the fact that $M(-f) = M^*(f)$. For lower sideband, we can invert the initial filter and pack a second message into the other (filtered) part of the spectrum.





What happens if we modulated with a square wave instead of a cosine wave? The square wave can be expressed as a Fourier series of cosines at odd harmonics of the fundamental frequency. In the frequency domain, this looks just like the coherent AM case but with copies of the message at all the harmonics (impulses at $f_c, 3f_c, 5f_c, \ldots$).

2.2.2 FM

Let

$$r(t) = \sin(\Theta(t)) = \sin\left(2\pi f_c t + \beta \int_{-\infty}^{t} m(\tau)\, d\tau\right)$$

This is a constant amplitude signal with changing frequency.

Instantaneous Frequency:

$$\frac{d\Theta(t)}{dt} = \omega(t)$$

What happens if we differentiate $r(t)$?

$$\frac{dr}{dt} = (2\pi f_c + \beta m(t)) \cos\left(2\pi f_c t + \beta \int_{-\infty}^{t} m(\tau)\, d\tau\right)$$

Since $f_c \gg m(t)$ and we assume β is relatively small (hence $\beta \int_{-\infty}^{t} m(\tau)\, d\tau \lll 1$) we can just throw this into an envelope detector and HPF (get rid of the DC term) to recover $\beta m(t)$ approximately.

$$r(t) \longrightarrow \boxed{d/dt} \longrightarrow \boxed{\begin{array}{c}\text{Envelope}\\ \text{Detector}\end{array}} \longrightarrow \boxed{\text{HPF}} \longrightarrow \beta m(t)$$

In particular,

$$\begin{aligned} r(t) &= \sin(2\pi f_c t)\cos(\beta\Theta(t)) + \cos(2\pi f_c t)\sin(\beta\Theta(t)) \\ &\approx (\cos 2\pi f_c t)\left(\beta \int_{-\infty}^{t} m(\tau)\, d\tau\right) + \sin 2\pi f_c t \qquad (\sin x = x) \end{aligned}$$

2.3 Feb 10 - Quiz 1 Review

What is a linear system?

$$L[au + bv] = aL[u] + bL[v], \quad a, b \in \mathbb{R}$$

Is $y = ax + b$ $(a, b \neq 0)$ linear? No. It doesn't satisfy scaling:

$$y = a(cx) + b \neq c(ax + b)$$

What is a linear time-invariant (LTI) system?

$$y(t) = L[x(t)] \implies y(t - t_0) = Lx(t - t_0)$$

What is a delta function?

1. $\int \delta(t)\, dt = 1$
2. $\delta(t) = 0$ for $t \neq 0$
3. $\int f(t)\delta(t - t_0)\, dt = f(t_0)$ (sifting)

Transfer function: $H(f) = \mathcal{F}(h(t))$ where $h(t)$ is the **impulse response** of the system.

$$\begin{matrix} \delta(t) \\ \delta(t-\tau) \end{matrix} \quad \text{---}\boxed{h(t)}\text{---} \quad \begin{matrix} y(t) \\ y(t-\tau) \end{matrix}$$

where the box represents a **convolution**:

$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)\, d\tau$$

(in fact, in the frequency domain, this operation is defined by $Y(f) = X(f)H(f)$)

Fourier Transform:

$$\mathcal{F}[x(t)] = \int x(t)e^{-j\,2\pi f t}\, dt = X(f)$$

Inverse Fourier:

$$\mathcal{F}^{-1}[X(f)] = \int X(f)e^{j\,2\pi f t}\, df = x(t)$$

Standard Fourier pairs:

1. $\mathcal{F}[\delta(t)] = 1$
2. $\mathcal{F}[e^{-j\,2\pi f_0 t}] = \delta(f + f_0)$
3. $\mathcal{F}[\delta(t - t_0)] = e^{-j\,2\pi f t_0}$
4. $\mathcal{F}[\cos 2\pi f_0 t] = \frac{1}{2}[\delta(f + f_0) + \delta(f - f_0)]$
5. $\mathcal{F}[\sin 2\pi f_0 t] = \frac{1}{2j}[\delta(f + f_0) - \delta(f - f_0)]$
6. $\mathcal{F}[\mathbb{1}_{x \in [-T/2, T/2]}] = \mathrm{sinc}(2\pi T f)$

Fourier Properties:

1. $\mathcal{F}[x(t - t_0)] = e^{j\,2\pi f t_0} X(f)$
2. $\mathcal{F}[ax(t)] = aX(f/a)$

3. $\mathcal{F}[x'(t)] = X(f) \cdot j\ 2\pi f$
4. $\mathcal{F}[\int_{-\infty}^{t} x(u)\ du] = \frac{X(f)}{j\ 2\pi f} + \frac{X(0)}{2}\delta(f)$
5. $\int_{-\infty}^{\infty} |x(t)|^2 = \int_{-\infty}^{\infty} |X(f)|^2$ (Parseval's)
6. $\mathcal{F}[X(t)] = x(-f)$

Useful tricks:

1. $|X(f)|^2 = X(f)X^*(f) = X(f)X(-f)$
2. $\cos(2\pi ft) = \frac{e^{j\ 2\pi ft} + e^{-j\ 2\pi ft}}{2}$
3. $\sin(2\pi ft) = \frac{e^{j\ 2\pi ft} - e^{-j\ 2\pi ft}}{2j}$
4. Fourier of odd function is purely imaginary, Fourier of even function is purely real.
5. δ is an even function: $\delta(-t) = \delta(t)$
6. $u(t) = \mathbb{1}_{t \geq 0}(t)$
7. $\sin(x + y) = \sin x \cos y + \cos x \sin y$
8. $\cos(x + y) = \cos x \cos y - \sin x \sin y$
9. $\sin^2 x = (1 - \cos 2x)/2$
10. $\cos^2 x = (1 + \cos 2x)/2$
11. $\cos\theta(t) = \sum_{m=0}^{\infty} \frac{(\theta(t))^{2m}}{(2m)!}$
12. $\sin\theta(t) = \sum_{m=0}^{\infty} (-1)^m \frac{(\theta(t))^{2m+1}}{(2m+1)!}$

Baseband: unmodulated signal (i.e. the message signal).

Passband: modulated signal (i.e. the transmitted signal).

2.3.1 Modulation

AM Coherent modulation: $r(t) = m(t)\cos 2\pi f_c t$, demodulate with $\cos 2\pi f_c t$ and LPF to recover $m(t)/2$.

AM Large carrier: $r(t) = (m(t) + A)\cos 2\pi f_c t$ $(A \gg m(t))$, demodulate with an envelope detector, LPF, and HPF to recover $m(t)$.

AM Single sideband: $r(t) = m(t)\cos 2\pi f_c t$ with an LPF applied to $m(t)$ before modulation (takes advantage of symmetry from $M(-f) = M^*(f)$ to use half the bandwidth), demodulate with $\cos 2\pi f_c t$ and LPF to recover $m(t)/2$.

Frequency modulation: $r(t) = \sin(2\pi f_c t + \beta \int_{-\infty}^{t} m(\tau)\ d\tau)$, demodulate by differentiating, envelope detecting, and HPF to recover $\beta m(t)$.

Phase modulation: $r(t) = \sin(2\pi f_c t + \beta m(t))$ (filter and integrate)

How do you determine the bandwidth of an FM signal? What does it look like in the frequency domain? Hint: use linear system properties

2.4 Feb 19 - Sampling

So far $m(t)$ has been a continuous-time signal. How do we represent it in the digital regime?

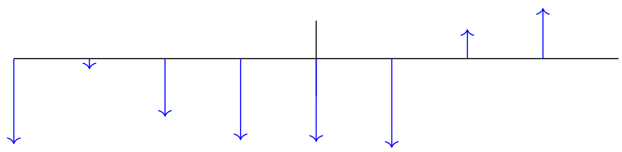
We assumed the spectrum of $m(t)$ is bandlimited to B Hz. How do we represent

$$m(t) \to \{m_1, m_2, \dots\}$$

a set of samples? (And especially a set of samples that we can use to reconstruct $m(t)$?)

Trick:

$$m(t)\lambda(t) = m(t) \sum_k \delta(t - kt)$$



What is the spectrum of this sampled signal?

$$\begin{aligned} M(f) \star \Lambda(f) &= M(f) \star \int e^{-j\,2\pi f t} \sum_k \delta(t - kT)\, dt \\ &= M(f) \star \sum_k e^{-j\,2\pi f k T} \end{aligned}$$

Unfortunately, this tells us nothing.

Instead, let's represent λ via **Fourier series**:

$$\lambda(t) = \sum_{\ell=-\infty}^{\infty} \lambda_\ell e^{j\,\frac{2\pi}{T}\ell t}$$

where

$$\lambda_\ell = \frac{1}{T}\int_{-T/2}^{T/2} \lambda(t) e^{-j\,\frac{2\pi}{T}\ell T}\, dt = \frac{1}{T}$$

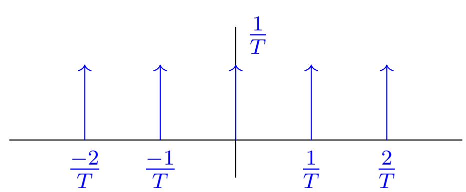
(note that on $t \in [-T/2, T/2]$, λ is a single impulse)

Hence,

$$\lambda(t) = \frac{1}{T}\sum_\ell e^{j\,\frac{2\pi}{T}\ell t}$$

and using Fourier pairs,

$$\Lambda(f) = \frac{1}{T}\sum_\ell \delta(f - \frac{\ell}{T})$$



Hence, we can draw $M(f) \star \Lambda(f)$

and using an LPF brings back our signal.

This also gives us the **Nyquist Criterion**

$$\frac{1}{T} - B > B \implies \boxed{\frac{1}{T} > 2B}$$

which tells us how fast we need to sample to be able to reconstruct the signal.

Remark: we could also prove this asymptotically with random samples

What exactly is happening with the LPF here?

$$\lambda(t) = \sum_{\ell} m(t)\delta(t - \ell T) = \sum \underbrace{m(\ell T)}_{m_\ell} \delta(t - \ell T)$$

What is the inverse fourier transform of $\mathbb{1}_{[-B,B]}$?

$$\int_{-B}^{B} e^{j\,2\pi f t}\, df = \frac{1}{j\,2\pi f t} e^{j\,2\pi f t}\Big|_{f=-B}^{f=B} = \frac{\sin 2\pi B t}{\pi t}$$

So the effect of the LPF in time-domain is

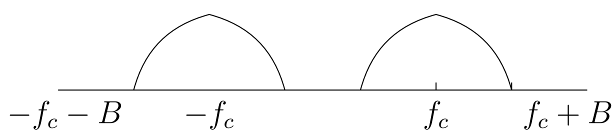
$$\frac{\sin 2\pi B t}{\pi t} \star \lambda(t) = \frac{\sin 2\pi B t}{\pi t} \star \sum_{\ell} m_\ell \delta(t - \ell T)$$

but anything convolved with an impulse is just the signal shifted so

$$\sum_{\ell} m_\ell \frac{\sin 2\pi B(t - \ell T)}{\pi (t - \ell T)}$$

which is our **interpellation formula**.

Consider this signal:



What's the lowest sampling rate we can use to reconstruct this signal?

Certainly, via Nyquist, we need to have $1/T > 2B$. Can we do better?

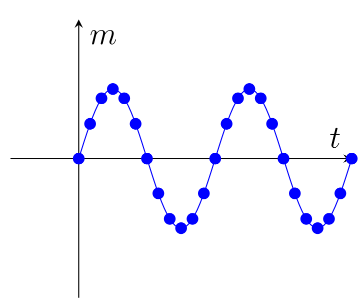
Using SSB, we can cut down to $1/T > 2f_c$ (since we only need to sample the sideband).

2.5 Feb 24 - Quantization

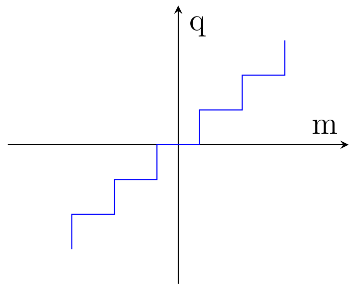
Motivated by the digital domain, we want to represent $m(t)$ as a collection of bits.

Using Nyquist, we sample

$$m(t) \to \{m(t_1), m(t_2), \ldots, m(t_{N-1})\}$$

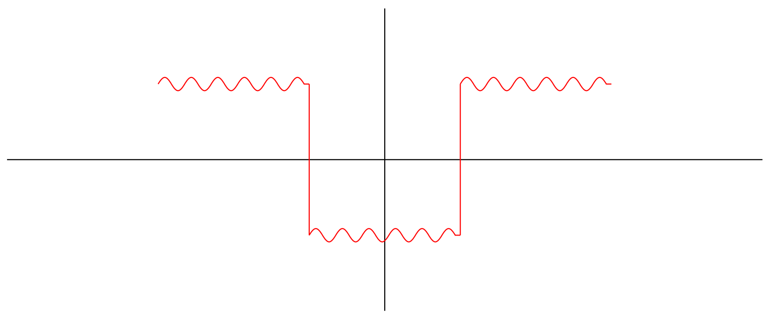


We will introduce a **N-level uniform quantizer** $Q(m)$ which maps $m(t_i)$ to the closest quantization level q_j for $j = 0, \ldots, N-1$.



Now, at the receiver we receive a sequence of quantization levels (say $\{q_3, q_1, q_7, q_2, \ldots\}$) and we can reconstruct $\widehat{m}(t)$ by using the interpellation formula on the quantized samples.

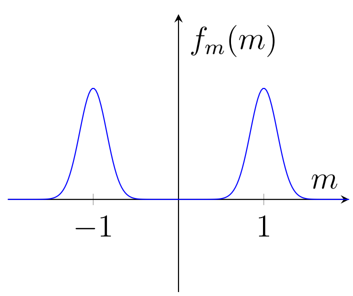
What if $m(t)$ looks like this?



Now all the detail is in the high frequencies so uniform quantization is not good!

We could try to make the quantization levels smaller but this wastes bits.

Another option is to tailor the quantizer given knowledge of the signal. Obviously, we can't know the signal exactly at the receiver to design a perfect quantizer but we can use some information about the distribution f_m of $m(t)$ to do better.



If we sample to see the $\mathbb{P}(M < m)$ for various levels m, we get the **cumulative distribution function** (CDF) of M, $F_M(m)$, and in fact

$$f_m(m) = \frac{d}{dm} F_m(m)$$

To review, our scheme looks like

$$\boxed{m(t)} \longrightarrow \boxed{\text{quantizer}} \longrightarrow \boxed{\text{transmit}} \sim\sim\sim \boxed{\text{receiver}} \longrightarrow \boxed{\widehat{m}(t)}$$

though, of course, $m(t) \neq \widehat{m}(t)$ because of loss and quantization error.

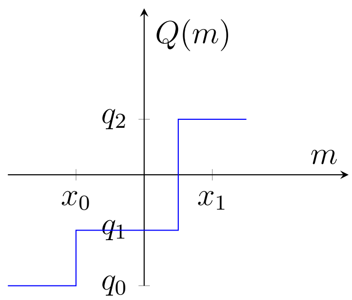
Let $Q(m)$ be the quantization function for some m sampled from $m(t)$.

Then

$$e^2 = (m - Q(m))^2$$
$$\overline{e^2} = \int f_M(m)(m - Q(m))^2 \, dm$$

is the **expected error**

What are the parameters that define a quantizer? Clearly the quantization levels $q_0, \ldots, q_{N-1}$ but also the **decision boundaries** $x_0, \ldots$



$$\mathcal{Q} = \{q_0, q_1, q_2, x_0, x_1\}$$

and we want to find

$$\min_{\mathcal{Q}} e^2$$

so we take partials, set to zero, solve, and check the second partials for convexity

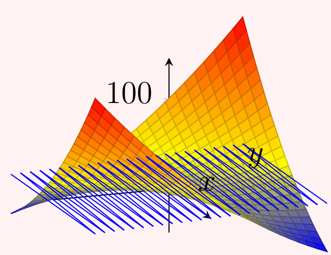
Exercise:

$$z = x^2 + y^2$$

$$\frac{\partial z}{\partial x} = 2x = 0$$
$$\frac{\partial z}{\partial y} = 2y = 0$$
$$\frac{\partial^2 z}{\partial x^2} = 2 > 0$$
$$\frac{\partial^2 z}{\partial y^2} = 2 > 0$$

So convex!

$$z = x^2 + 4xy + y^2$$



$$
\begin{aligned}
\frac{\partial z}{\partial x} &= 2x + 4y = 0 \\
\frac{\partial z}{\partial y} &= 2y + 4x = 0 \\
\frac{\partial^2 z}{\partial x^2} &= 2 \\
\frac{\partial^2 z}{\partial y^2} &= 2
\end{aligned}
$$

But this does not have a global minimum!

Let's try this for our error function.

$$
\begin{aligned}
e^2 &= \int (m - Q(m))^2 f_M(m)\; dm \\
&= \int_{-\infty}^{x_0} (m - q_0)^2 f_M(m)\; dm + \int_{x_0}^{x_1} (m - q_1)^2 f_M(m)\; dm + \cdots + \int_{x_{N-1}}^{\infty} (m - q_{N-1})^2 f_M(m)\; dm \\
\frac{\partial e^2}{\partial q_i} &= \frac{\partial}{\partial q_i} \int_{x_{i-1}}^{x_i} 2(q_i - m) f_M(m)\; dm \\
&= 2 \int_{x_{i-1}}^{x_i} (m - q_i) f_M(m)\; dm = 0 \\
\frac{\partial e^2}{\partial x_i} &= \frac{\partial}{\partial x_i} \left[\int_{x_{i-1}}^{x_i} (m - q_i)^2 f_M(m)\; dm + \int_{x_i}^{x_{i+1}} (m - q_{i+1})^2 f_M(m)\; dm \right] \\
&= (m - q_i)^2 f_M(x_i) - (m - q_{i+1})^2 f_M(x_i) = 0
\end{aligned}
$$

via

Leibniz's Rule:

$$
\frac{d}{dx} \int_{a(x)}^{b(x)} f(x, y)\; dy = f(x, b(x)) \cdot \frac{db}{dx} - f(x, a(x)) \cdot \frac{da}{dx} + \int_{a(x)}^{b(x)} \frac{\partial f}{\partial x}\; dy
$$

which implies

$$
\begin{aligned}
(x_i - q_i)^2 &= (x_i - q_{i+1})^2 \\
x_i^2 - 2x_i q_i + q_i^2 &= x_i^2 - 2x_i q_{i+1} + q_{i+1}^2
\end{aligned}
$$

$$
\boxed{q_i = \frac{q_{i-1} + q_{i+1}}{2}}
$$

and

$$
\int_{x_{i-1}}^{x_i} m f_M(m)\; dm = q_i \underbrace{\int_{x_{i-1}}^{x_i} f_M(m)\; dm}_{p_m = \mathbb{P}(m \in (x_{i-1}, x_i))}
$$

(assuming $f_M(x_i) \neq 0$)

At last giving an explicit formula for the quantization levels and decision boundaries:

$$
\boxed{q_i = \int_{x_{i-1}}^{x_i} m \frac{f_M(m)}{p_m}\; dm}
$$

which is precisely the *conditional average on M* given $M \in (x_{i-1}, x_i)$

Together, these are exactly the **Lloyd-Max Conditions** for optimal quantization.

2.6 Feb 26 - Lloyd Max Review

Recall: we sample $m(t) \to \{m_1, m_2, \dots\}$ creating a pdf $f_M(m)$. The intuition is we want to use information from this pdf to design a quantizer that minimizes the expected error while using a reasonable number of quantization levels.

Define an *error function*

$$e^2 = \int (m - Q(m))^2 f_M(m)\, dm$$

for $Q(m)$ a quantization function defined by quantization levels $q_0, \dots, q_{N-1}$ and breakpoints $x_0, \dots, x_{N-2}$.

Since Q is piecewise constant, we can write

$$e^2 = \int_{-\infty}^{x_0} (m - Q(m))^2 f_M(m)\, dm + \int_{x_0}^{x_1} (m - Q(M))^2 f_M(m)\, dm + \cdots + \int_{x_{N-2}}^{\infty} (m - Q(m))^2 f_M(m)\, dm$$

We will blindly take partials using Leibniz and set to zero to minimize.

$$\begin{aligned} \frac{\partial e^2}{\partial x_i} &= \frac{\partial}{\partial x_i}\left[\int_{x_{i-1}}^{x_i} (m - q_{i+1})^2 f_M(m)\, dm + \int_{x_{i-1}}^{x_i} (m - q_i)^2 f_M(m)\, dm\right] \\ &= -(x_i - q_{i+1})^2 f_M(x_i) + (x_i - q_i)^2 f_M(x_i) = 0 \end{aligned}$$

Assuming $f_M(x_i) \neq 0$, we get

$$\begin{aligned} (x_i - q_i)^2 &= (x_i - q_{i+1})^2 \\ x_i^2 - 2x_i q_i + q_i^2 &= x_i^2 - 2x_i q_{i+1} + q_{i+1}^2 \\ 2(q_{i+1} - q_i)x_i &= q_{i+1}^2 - q_i^2 \\ 2(q_{i+1} - q_i)x_i &= (q_{i+1} - q_i)(q_{i+1} + q_i) \end{aligned}$$

so

$$\boxed{x_i = \frac{q_{i+1} + q_i}{2}}$$

that is, the breakpoints are the midpoints of the quantization levels - a very nice result!

Now let's go back and take the partials with respect to q_i. First let $p_i = \mathbb{P}(m \in (x_{i-1}, x_i))$.

$$\begin{aligned} \frac{\partial e^2}{\partial q_i} &= \frac{\partial}{\partial q_i} \int_{x_{i-1}}^{x_i} 2(q_i - m) f_M(m)\, dm \\ &= 2\int_{x_{i-1}}^{x_i} (m - q_i) f_M(m)\, dm = 0 \\ &= \int_{x_{i-1}}^{x_i} m f_M(m)\, dm - q_i \int_{x_{i-1}}^{x_i} f_M(m)\, dm = 0 \end{aligned}$$

so

$$\int_{x_{i-1}}^{x_i} m f_M(m)\, dm = q_i \int_{x_{i-1}}^{x_i} f_M(m)\, dm$$

hence

$$\boxed{q_i = \int_{x_{i-1}}^{x_i} m \frac{f_M(m)}{p_i}\, dm = \mathbb{E}[M \mid m \in (x_{i-1}, x_i)]}$$

Together, the boxed equations are the **Lloyd-Max Conditions** for optimal quantization.

If given $\{x_i\}$, the optimal $\{q_i\}$ are given by the conditions. Similarly, if given $\{q_i\}$, the optimal $\{x_i\}$ are given by the conditions.

In particular, this optimality ensures the quantizer will be monotonic since it intuitively should correspond to "binning" the amplitudes of the samples.

> **Exercise:** Prove that the Lloyd-Max conditions are optimal (Hint: see quantization notes on canvas)

As we will see, our error function is *convex*.

Convexity (according to high school calculus): $f(x)$ convex iff $\frac{d^2 f(x)}{dx^2} > 0 \quad \forall x$

Convexity (for adults): $f(x_1, \ldots, x_n)$ convex iff $\forall x_1, x_2 \in \mathbb{R}^n$ and $\lambda \in [0, 1]$,

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda) f(x_2)$$

(corresponding to a chord between any two points of f being above the curve)

Strictly convex: f is strictly convex if the above inequality is strict for $\lambda \in (0, 1)$

> **Exercise:** Show that the two convexity definitions are equivalent. (Hint: use the Hessian)

Chapter 3

Interlude - Introduction to Probability and Stochastic Processes

3.1 March 3

Random variable: a map $X : \Omega \to \mathbb{R}$ for Ω an *event space.*

This map can take single events or sets of events as input. (Imagine for example the probability of rolling a 2 on two dice (one possibility) vs the probability of rolling a 7 on at least one die (multiple possibilities))

Probability Mass Function (pmf): $\{\omega_1, \ldots, \omega_n\} \to \{p_1, \ldots, p_n\}$ where $p_i = \mathbb{P}(\omega_i)$ satisfying

1. $\sum_{i=1}^n p_i = 1$.
2. $p_i \geq 0$ for all i.

Cumulative distribution function (cdf): $P(x) = \mathbb{P}(X \leq x)$ for a random variable X.

Expectation: $\mathbb{E}[X] = \sum_{i=1}^n p_i X(\omega_i) := \overline{X}$

Examples:

- $\mathbb{E}[X^2] = \sum_{i=1}^n p_i X(\omega_i)^2$
- $\mathbb{E}[f(X)] = \sum_{i=1}^n p_i f(X(\omega_i))$

Property: Expectation is linear:

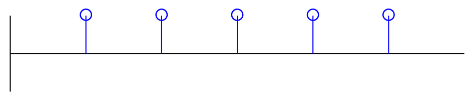
$$\mathbb{E}[f(X) + g(Y)] = \sum p_n(f(X) + g(Y)) = \sum p_n f(X) + \sum p_n g(Y) = \mathbb{E}[f(X)] + \mathbb{E}[g(Y)]$$

Exercise: Calculate $\mathbb{E}[(X-\overline{X})^2]$

$$\begin{aligned}
\mathbb{E}[(X-\overline{X})^2] &= \sum_{i=1}^{n} p_i (X(\omega_i)-\overline{X})^2 \\
&= \sum_{i=1}^{n} p_i (X(\omega_i)^2 - 2X(\omega_i)\overline{X} + \overline{X}^2) \\
&= \sum_i p_i X_i^2 - 2\overline{X}\sum p_i X_i + \overline{X}\sum p_i \\
&= \mathbb{E}[X^2] - 2\overline{X}\cdot\overline{X} + \overline{X}^2 \\
&= \mathbb{E}[X^2] - \overline{X}^2 \\
&= \mathrm{Var}\,[X]
\end{aligned}$$

Examples of PMFs:

1. Uniform



2. Poisson

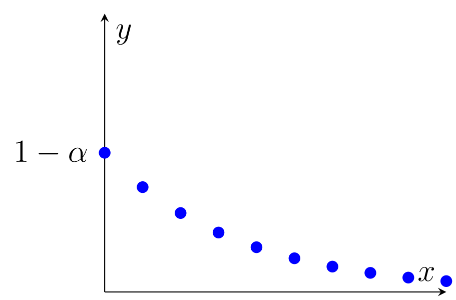
$$p_\lambda(k) = \frac{(\lambda T)^k}{k!} e^{-\lambda T}, \quad k = 0, 1, 2, \ldots$$

(nicely satisfying $\mathbb{E}[K] = \lambda T$)

3. Geometric

$$p_\alpha(k) = (1-\alpha)\alpha^k, \quad \alpha \in [0,1],\ k = 0, 1, 2, \ldots$$

(recall $\sum_{k=0}^{\infty} \alpha^k = 1/(1-\alpha)$)



4. Binomial

$$p_\alpha(k) = \binom{N}{k} \alpha^k (1-\alpha)^{N-k}$$

for k the number of heads in N Bernoulli trial with $\mathbb{P}(H) = \alpha$ and $\mathbb{P}(T) = 1-\alpha$

(notice $\mathbb{E}[K] = \alpha K$)

Now let's move on to **continuous random variables** where we have a **probability density function (pdf)** $f_X(x)$ such that

1. $f_X(x) \geq 0$ for all x.
2. $\int_{-\infty}^{\infty} f_X(x)\, dx = 1$

and a **cumulative distribution function (CDF)**

$$F_X(x) = \int_{-\infty}^{x} f_X(\xi)\, d\xi = \mathbb{P}(X \leq x)$$

Expection of a continuous RV:

$$\mathbb{E}[X] = \int x f_X(x)\, dx := \overline{X}$$

Variance of a continuous RV:

$$\mathrm{Var}\,[X] = \mathbb{E}[(X - \overline{X})^2] = \int (x - \overline{X})^2 f_X(x)\, dx = \mathbb{E}[X^2] - \mathbb{E}[\overline{X}]^2 = \mathbb{E}[X^2] - \overline{X}^2 = \sigma_X^2$$

Notice that from the definition of the CDF, we have

$$F_X(x) = \int_{-\infty}^{x} f_X(y)\, dy \iff f_X(x) = \frac{d}{dx} F_X(x)$$

Joint Random Vairables: Let X, Y be two random vairables with joint pdf $f_{X,Y}(x, y)$ satisfying

1. $\int\int f_{X,Y}(x, y)\, dx\, dy = 1$
2. $f_{X,Y}(x, y) \geq 0$ for all x, y.

notice that this first property leads to the **marginal pdfs**

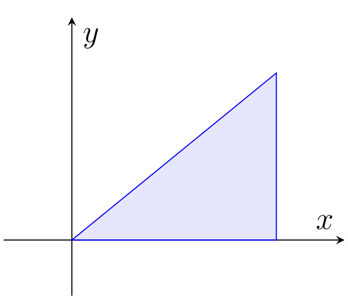
$$f_X(x) = \int f_{X,Y}(x, y)\, dy$$
$$f_Y(y) = \int f_{X,Y}(x, y)\, dx$$

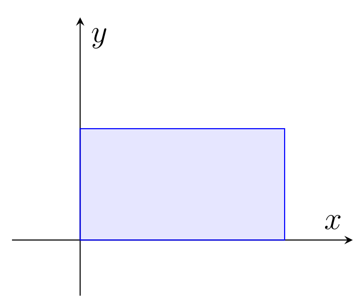
hence

$$\mathbb{E}[X] = \overline{X} = \int\int x f_{X,Y}(x, y)\, dx\, dy$$
$$\mathbb{E}[Y] = \overline{Y}$$

Notice that we do not know that X and Y are independent.

Consider these two examples:





If we know $X \cdot Y = 1$, then in the first example, knowing $X = 0$ fixes Y. In the second case, knowing $X = 0$ only tells us that $Y \sim \mathrm{Unif}(0, 1)$ – no new information.

Conditional probability:

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

(the joint pdf divided by the marginal pdf of Y)

Bayes' Theorem:

$$f_{X,Y}(x,y) = f_{X|Y}(x \mid y) f_Y(y) = f_{Y|X}(y \mid x) f_X(x)$$

Independence: X and Y are independent iff

$$f_{X|Y}(x \mid y) = f_X(x)$$

Intuitively,, telling me y tells me nothing about x.

Exercise: Show that for independent X, Y

$$f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

Correlation:

$$\mathbb{E}[XY] = \int\int x\, y\; f_{X,Y}(x,y)\; dx\, dy$$

which is precisely $\mathbb{E}[X] \cdot \mathbb{E}[Y]$ for independent X, Y

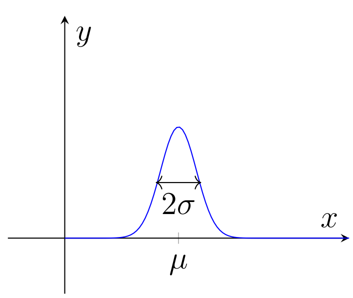
Covariance:

$$\mathbb{E}[(X - \overline{X})(Y - \overline{Y})] = \int\int (X - \overline{X})(Y - \overline{Y}) f_{X,Y}(x,y)\; dx\, dy$$

and since $\mathbb{E}[X - \overline{X}] = 0$, we have that the covariance is *zero* for independent X, Y

Gaussian:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



Joint Gaussian:

$$f_X(x_1, \ldots, x_N) = \left(\frac{1}{2\pi}\right)^{N/2} |K|^{-1/2} \exp\left(\frac{-(\vec{x} - \vec{\mu})^\top K^{-1} (\vec{x} - \vec{\mu})}{2}\right)$$

for covariance matrix

$$K = \mathbb{E}[(\vec{x} - \vec{\mu})(\vec{x} - \vec{\mu})^\top] = \begin{pmatrix} \sigma_1^2 & \mathrm{Cov}\,[X_1, X_2] & \ldots & \mathrm{Cov}\,[X_1, X_N] \\ \mathrm{Cov}\,[X_2, X_1] & \sigma_2^2 & \ldots & \mathrm{Cov}\,[X_2, X_N] \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}\,[X_N, X_1] & \mathrm{Cov}\,[X_N, X_2] & \ldots & \sigma_N^2 \end{pmatrix}$$

Theorem: If X, Y are jointly Gaussian,

$$\mathrm{Cov}\,(X, Y) = 0 \implies X \perp Y$$

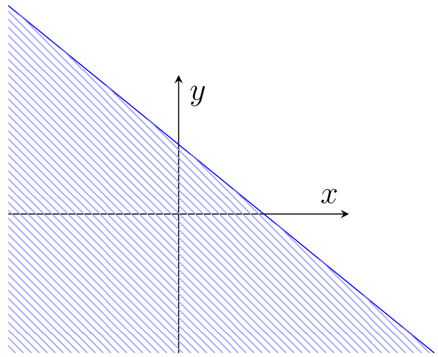
and for any X, Y independent

$$X \perp Y \implies \mathrm{Cov}\,(X, Y) = 0$$

Adding Random Variables: Suppose $Z = X + Y$. Given $f_{X,Y}(x, y)$, what is $F_Z(z)$?

First,

$$\mathbb{P}(Z \le z) = \mathbb{P}(X + Y \le z)$$

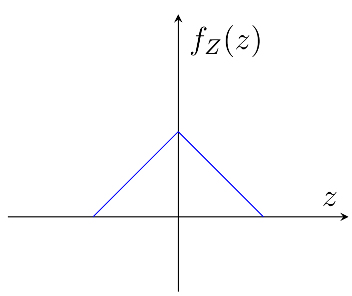


Hence

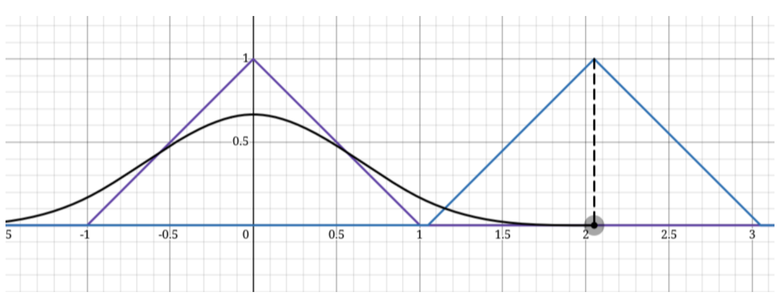
$$\begin{aligned}
\mathbb{P}(Z \le z) &= \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y)\, dy\, dx \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_X(x) f_Y(y)\, dy\, dx \qquad (X \perp Y) \\
\frac{d}{dz}\mathbb{P}(Z \le z) &= \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z - x)\, dx \qquad \text{(Leibniz)} \\
&= (f_X \star f_Y)(z)
\end{aligned}$$

Convolution!!

Example, if $Z = X + Y$ for $X, Y \overset{\text{iid}}{\sim} \text{Unif}(-1, 1)$, then



We have seen this over an over. Now what happens if we take $Z + Z$?



We find that it smooths out. What happens if we keep iterating?

Central limit theorem: For X_k iid,

$$\lim_{n \to \infty} \frac{1}{\sqrt{n}} \sum_{k=1}^{n} (X_k - \overline{X}) \to \mathcal{N}(0, \sigma^2)$$

3.2 March 5 - Gaussians and Stochastic Processes

3.2.1 Gaussians

Univariate Gaussian:

$$f_{\mu,\sigma}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

Multivariate Gaussian:

$$f_{K,\vec{m}u} = \left(\frac{1}{2\pi}\right)^{N/2} |K|^{-1/2} \exp\left(-\frac{1}{2}(\vec{x}-\vec{\mu})^\top K^{-1}(\vec{x}-\vec{\mu})\right)$$

Consider the case where $\vec{X} = (X_1, X_2)^\top$. What do we need to know to find the joint Gaussian pdf? Only the mean $\vec{\mu}$ and the covariance matrix K.

$$K = \mathbb{E}\begin{bmatrix} \operatorname{Var} X_1 & \operatorname{Cov}[X_1, X_2] \\ \operatorname{Cov}[X_2, X_1] & \operatorname{Var} X_2 \end{bmatrix} = \mathbb{E}\begin{bmatrix} (X_1-\mu_1)^2 & (X_1-\mu_1)(X_2-\mu_2) \\ (X_2-\mu_2)(X_1-\mu_1) & (X_2-\mu_2)^2 \end{bmatrix} = \mathbb{E}[(\vec{X}-\vec{\mu})^\top(\vec{x}-\vec{\mu})]$$

Notation: $\operatorname{Var} X_1 = \sigma_1^2$ and $\operatorname{Cov}[X_1, X_2] = \rho_{12}$

What if $\rho_{12} = \rho_{21} = 0$? Then

$$K = \begin{pmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{pmatrix}$$

so

$$K^{-1} = \begin{pmatrix} 1/\sigma_1^2 & 0 \\ 0 & 1/\sigma_2^2 \end{pmatrix}, \qquad |K| = \sigma_1^2\sigma_2^2$$

hence

$$\begin{aligned} f_{K,\vec{\mu}}(x) &= \left(\frac{1}{2\pi}\right) \frac{1}{\sqrt{\sigma_1^2\sigma_2^2}} \exp\left(-\frac{1}{2}\begin{pmatrix} x_1-\mu_1 \\ x_2-\mu_2 \end{pmatrix}^\top \begin{pmatrix} 1/\sigma_1^2 & 0 \\ 0 & 1/\sigma_2^2 \end{pmatrix}\begin{pmatrix} x_1-\mu_1 \\ x_2-\mu_2 \end{pmatrix}\right) \\ &= \left(\frac{1}{2\pi}\right) \frac{1}{\sqrt{\sigma_1^2\sigma_2^2}} \exp\left(-\frac{1}{2}\left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]\right) \\ &= \frac{1}{\sqrt{2\pi\sigma_1^2}} e^{-(x_1-\mu_1)^2/(2\sigma_1^2)} \cdot \frac{1}{\sqrt{2\pi\sigma_2^2}} e^{-(x_2-\mu_2)^2/(2\sigma_2^2)} \\ &= f_{\mu_1,\sigma_1}(x_1) \cdot f_{\mu_2,\sigma_2}(x_2) \end{aligned}$$

Proposition: If $\vec{X}$ is a set of jointly Gaussian RVs and all their covariances are zero, they are independent.

As we will see, joint Gaussian RVs have some very nice properties.

Let $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ be independent (*Notation:* $X_1 \perp X_2$). What is the distribution of $Z = X_1 + X_2$?

Lemma: For X, Y independent and $Z = X + Y$, $\operatorname{Var} Z = \operatorname{Var} X + \operatorname{Var} Y$

Proof:

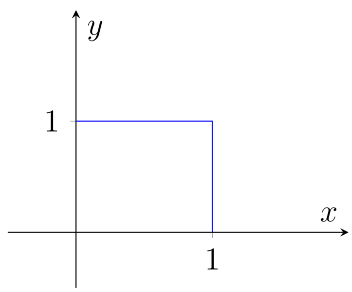
$$\begin{aligned}
\mathbb{E}[Z^2] &= \mathbb{E}[(X - \overline{X} + Y - \overline{Y})^2] \\
&= \mathbb{E}[(X - \overline{X})^2] + \mathbb{E}[(Y - \overline{Y})^2] + 2\mathbb{E}[(X - \overline{X})(Y - \overline{Y})] \\
&= \operatorname{Var} X + \operatorname{Var} Y + 2 \underbrace{\operatorname{Cov}(X, Y)}_{=0} \\
&= \sigma_1^2 + \sigma_2^2
\end{aligned}$$

Exercise: Prove that $Z = X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

Proposition: Let $\vec{X}, \vec{Y}$ be jointly Gaussian. Then for A, B linear operators,

1. $Z = A\vec{X}$ is jointly Gaussian
2. $Z = A\vec{X} + B\vec{Y}$ is jointly Gaussian

Let $X \sim \operatorname{Unif}([0, 1]) = u(x) - u(x - 1)$, for u the unit step.



Now let $Y = X^2$. What is the CDF of Y?

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X^2 \leq y) = \mathbb{P}(X \leq \sqrt{y}) = \sqrt{y}$$

hence

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{1}{2\sqrt{y}}$$

3.2.2 Stochastic Processes

Recall: A random variable X is a mapping from event space $X : \Omega \to \mathbb{R}$.

Random process: A random process is a mapping from event space to functions $X : \omega \mapsto f(t, \omega)$.

If we know t, this is just a random variable. If we know ω, this is a random function. If we know both, this is just a number.

Say we have a graph of Chris' position over time and we want to predict where he will be at time t to throw a dart at him. We can model his position as a random process $X(t)$.

What are some things we might want to know about this process?

- $\overline{X}(t) = \mathbb{E}[X(t)]$ - the expected position at time t
- $\operatorname{Var}[X(t)] = \mathbb{E}[X(t) - \overline{X}(t)]$ - the variance of his position at time t
- $\rho_X(t, \tau) = \mathbb{E}[X(t)X(t + \tau)]$ - the correlation of his position at time t and $t + \tau$ (how much does knowing his position at time t tell us about his position at time $t + \tau$?)

Remark: in this class, we will assume all processes are *stationary:* $\rho_X(t,\tau) = \rho(\tau)$ for all t.

What we would really like, though, is to know the **joint distribution** of $X(t_1), \ldots, X(t_n)$ for all n and $t_1, \ldots, t_n$ with n very large. In theory, this gives us full knowledge of the process. In practice, this is impossible to know, so we will make some assumptions about the process to make it tractable.

Stationary Gaussian Process: $\{X_t : t \in \mathbb{Z}\}$ is a stationary Gaussian process if

1. $\mathbb{E}[X(t)] = 0$
2. $\mathbb{E}\Big[\big(X(t) - \mathbb{E}[X(t)]\big)\big(X(t+\tau) - \mathbb{E}[X(t+\tau)]\big)\Big] = K(\tau)$

In particular, this means that if we know the mean and covariance function of a stationary Gaussian process, we know the full joint distribution of the process – precisely what we said was impossible to know in general.

White Gaussian Process: A stationary Gaussian process with

$$K_X(\tau) = N_0\delta(\tau)\mathbb{E}[X(t)X(t+\tau)]$$

for $N_0 \in \mathbb{R}$.

Why is it called white? Consider its **spectral density**

$$S_X(f) = \mathcal{F}[K_X(\tau)] = \mathcal{F}[N_0\delta(\tau)] = N_0$$

which is constant across all frequencies, hence the name white (like white light which contains all frequencies of visible light).

3.3 March 10

Recall: a jointly Gaussian distribution is given by

$$\left(\frac{1}{2\pi}\right)^{N/2} |K|^{-1/2} \exp\left(-\frac{1}{2}(\vec{x}-\vec{\mu})^\top K^{-1}(\vec{x}-\vec{\mu})\right)$$

which requires only the mean and the covariance matrix to specify completely (first and second moments).

$$\begin{aligned} R_X(\tau) &= \mathbb{E}[X(t)X(t+\tau)] \\ K_X(\tau) &= \mathbb{E}[(X(t)-\mu)(X(t+\tau)-\mu)] \\ R_{XY}(\tau) &= \mathbb{E}[X(t)Y(t+\tau)] \end{aligned}$$

Last time, we started talking about stochastic processes because signals are random due to noise, interference, and other factors. Today, we will start thinking about *communications in the face of noise.*

For a random process, we need to know the joint distribution $f_{X(t_1),\ldots,X(t_n)}(x_1,\ldots,x_n)$ for all n and $t_1,\ldots,t_n$ to completely characterize it (even ignoring the fact that there are uncountably many t). This is HARD. But Gaussians are wonderful!

For a stationary Gaussian process, given m (constant because stationary) and $K(\tau)$, what is $f_{X(t_1),X(t_2)}(x_1,x_2)$?

In particular, what is the relationship between the covariance function $K(\tau)$ and the covariance matrix K?

$$K = \mathbb{E}\left[\begin{pmatrix} X(t_1) \\ X(t_2) \end{pmatrix}\begin{pmatrix} X(t_1) & X(t_2) \end{pmatrix}\right] = \begin{pmatrix} K(0) & K(t_1-t_2) \\ K(t_2-t_1) & K(0) \end{pmatrix}$$

(though we have real symmetry, so $K(t_1-t_2) = K(t_2-t_1)$)

Then, letting $m = 0$ for simplicity, we have

$$f_{X(t_1),X(t_2)}(x_1, x_2) = \frac{1}{2\pi} |K|^{-1/2} \exp\left(-\frac{1}{2}\vec{x}^\top K^{-1}\vec{x}\right)$$

Suppose we input a gaussian process into a linear time-invariant (LTI) system with impulse response $h(t)$. What is the output?

$$Y(t) = X(t) \star h(t) = \int_{-\infty}^{\infty} X(\tau)h(t-\tau)d\tau$$

for each t, τ this is just a linear combination of Gaussian RVs, hence $Y(t)$ is also a Gaussian process.

What is the mean of $Y(t)$? Let's assume $\mathbb{E}[X(t)] = 0$ and stationary for simplicity. Then

$$\mathbb{E}[Y(t)] = \int\int X(\tau)h(t-\tau)\, d\tau\, dt = \int h(t-\tau)\mathbb{E}[X(t)]\, d\tau = 0$$

What about the covariance? Changing the variables of integration,

$$\begin{aligned}
\mathbb{E}[Y(t)Y(t+\tau)] &= \mathbb{E}\left[\left(\int X(\lambda)h(t-\lambda)\, d\lambda\right)\left(\int X(\sigma)h(t+\tau-\sigma)\, d\sigma\right)\right] \\
&= \mathbb{E}\left[\int\int X(\lambda)X(\sigma)h(t-\lambda)h(t+\tau-\sigma)\, d\sigma d\lambda\right] \\
&= \mathbb{E}\left[\int\int K_X(\lambda-\sigma)h(t-\lambda)h(t+\tau-\sigma)\, d\lambda\, d\sigma\right]
\end{aligned}$$

since

$$K_X(\tau) = \mathbb{E}[X(t_1)X(t_2)] = \mathbb{E}[X(t_1)X(t_1 + \underbrace{(t_2 - t_1)}_{\tau})] = K_X(t_2 - t_1)$$

Now, if we further assume this is a white Gaussian process ($K_X(\lambda - \sigma) = N_0\delta(\lambda - \sigma)$), then

$$\begin{aligned}
\mathbb{E}[Y(t)Y(t+\tau)] &= \dots \\
&= \mathbb{E}\left[\int\int K_X(\lambda-\sigma)h(t-\lambda)h(t+\tau-\sigma)\, d\lambda\, d\sigma\right] \\
&= N_0 \int h(t-\sigma)h(t+\tau-\sigma)\, d\sigma \\
&= K_Y(\tau)
\end{aligned}$$

Is this actually a function of t? Yes. Consider $z = t - \sigma$ then for all t,

$$K_Y(\tau) = N_0 \int h(z)h(z+\tau)\, d\tau \implies \text{still stationary (only the offset changes)}$$

3.3.1 Signalling

Define a signalling interval $(0, T)$. We will send a single bit of information $b \in \{-1, 1\}$.

Sent through the channel, we receive $y(t)$ and want to do some magic to recover $\widehat{b}$ as accurately as possible.

We can model the channel as adding a stationary, white, zero-mean noise $w(t)$ (that is, $\mathbb{E}[w(t)w(t+\tau)] = N_0\delta(\tau)$) to the signal, so

$$y(t) = x(t) + w(t)$$

Our ability to decode the message then clearly depends on how large the noise w is.

Let's put $y(t)$ into an LTI with impulse response $h(t)$, outputting $z(t)$. We will sample $z(T)$ to decide if $\widehat{b} = -1$ or $\widehat{b} = 1$.

$$y(t) = bx(t) + w(t) \implies z(t) = \underbrace{b\int x(\tau)h(t-\tau)\,d\tau}_{\text{info}} + \int w(\tau)h(t+\tau)\,d\tau$$

so

$$z(T) = \int_0^T x(\tau)h(T-\tau)\,d\tau + \int_0^T w(\tau)h(T-\tau)\,d\tau$$

We know that both are zero in expectation so how can we measure the relative sizes of each?

Signal to noise ratio (SNR):

$$\begin{aligned}
\text{SNR} &= \frac{\mathbb{E}\left[\left(b\int_0^T x(\tau)h(T-\tau)\,d\tau\right)^2\right]}{\mathbb{E}\left[\left(\int_0^T w(\tau)h(T-\tau)\,d\tau\right)^2\right]} \\
&= \frac{\left(\int_0^T x(\tau)h(T-\tau)\,d\tau\right)^2 \mathbb{E}[b^2]}{\mathbb{E}\left[\int\int w(\tau)w(\lambda)h(T-\tau)w(T-\lambda)\,d\tau\,d\lambda\right]} \\
&= \frac{\left(\int_0^T x(\tau)h(T-\tau)\,d\tau\right)^2}{\int_0^T\int_0^T N_0\delta(\tau-\lambda)h(T-\tau)h(T-\lambda)\,d\tau\,d\lambda} \\
&= \frac{\left(\int_0^T x(\tau)h(T-\tau)\,d\tau\right)^2}{N_0\int_0^T h^2(T-\lambda)\,d\lambda}
\end{aligned}$$

where $\mathcal{E}_h = N_0\int_0^T h^2(T-\lambda)\,d\lambda$ is the **energy** of h (integral of a square!)

In particular, this means we want to find an h to maximize this ratio.

We know

$$\left(\int_0^T x(\tau)h(T-\tau)\,d\tau\right)^2 \le \left(\int_0^T x^2(\tau)\,d\tau\right)\left(\int_0^T h^2(T-\tau)\,d\tau\right)$$

by Cauchy-Schwarz.

Cauchy-Schwarz:

$$\left(\int x(t)y(t)\right)^2 \le \left(\int |x(t)|^2\,dt\right)\left(\int |y(t)|^2\,dt\right)$$

with equality iff $x(t) \propto y(t)$.

And since we want to maximize the numerator, we want equality. That is, the optimal h is precisely $h(t-\tau) = \alpha x(\tau)$ for $\alpha \in \mathbb{R}$. In which case,

$$\max \text{SNR} = \frac{\left(\int_0^T x^2(\tau)\,d\tau\right)\alpha^2\int_0^T x^2(\tau)\,d\tau}{N_0\alpha^2\int x^2(\tau)\,d\tau} = \frac{\int_0^T x^2(\tau)\,d\tau}{N_0} = \frac{\mathcal{E}_x}{N_0}$$

Hence, which LTI system maximizes SNR? Precisely the *matched filter* which scales a flipped version of x, resulting in an SNR equal to the quotient of the energy in x and the inherent noise level.

This means that we don't need to spend much time designing a careful x: we just want to maximize energy!

Recall

$$z(T) = \int_0^T x(\tau)h(T-\tau)\, d\tau + \int_0^T w(\tau)h(T-\tau)\, d\tau$$

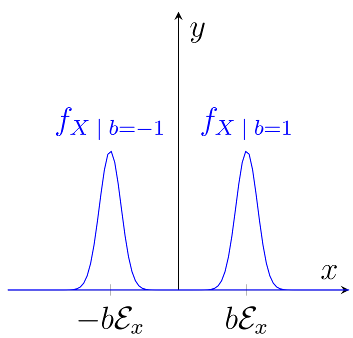
now with the matched filter $h(T-\tau) = x(\tau)$,

$$Z(T) = b\mathcal{E}_x + W$$

with $W = \int_0^T w(\tau)h(T-\tau)\, d\tau \sim \mathcal{N}(0, N_0\mathcal{E}_x)$

$$\mathbb{E}[W^2] = \mathbb{E}[\int_0^T \int_0^T w(\tau)w(\lambda)x(\tau)x(\lambda)\, d\tau\, d\lambda] = N_0\mathcal{E}_x$$

If $b = 1$, what does $Z(T)$ look like?



as we will see, if we sample at $Z(T)$, we will then be very likely to correctly decode b as 1 or -1 depending on the marginals.

What if the distributions are more closely overlapping or not so simple? This will be our focus moving forward.